

IT Troubleshooting Assessment

System Diagnostics & Resolution Procedures

1. Laptop Power Failure Diagnosis

Scenario: A user reports their laptop won't power on even when plugged in.

Here are 5 troubleshooting steps to diagnose the cause:

- 1. Verify the Power Source and Adapter:** Plug the AC adapter directly into a known working wall outlet (bypassing surge protectors). Inspect the power brick and cables for physical damage or fraying.
Checking for: Ensuring that the issue is not simply a faulty power strip or a dead AC adapter failing to deliver power to the machine.
- 2. Perform a Hard Reset (Power Drain):** Disconnect the AC adapter, remove the external battery (if applicable), and hold the power button down for 30-60 seconds. Reconnect the AC adapter and attempt to power on.
Checking for: Residual static charge ("flea power") built up on the motherboard components that is preventing the system from completing its boot sequence.
- 3. Inspect the DC Jack / Charging Port:** Examine the laptop's charging port with a flashlight for bent pins, debris, or a loose connection.
Checking for: Physical hardware damage that prevents the power adapter from making a solid electrical connection with the motherboard.
- 4. Check System LED Indicators:** Plug in the adapter and look for battery charging lights or power indicator LEDs. Press the power button and observe if any lights blink (and count the blink code) or if the fan spins briefly.
Checking for: Diagnostic error codes. If LEDs light up, the motherboard is receiving power. If they blink in a specific pattern, it indicates a specific hardware failure (e.g., RAM or CPU). If completely dark, the motherboard may be dead.
- 5. Test Without the Battery:** If the battery is removable, take it out and try to power the laptop using only the AC adapter.
Checking for: A short-circuited or completely dead battery that is drawing all the power and preventing the motherboard from POSTing.

2. Blank Monitor Troubleshooting Flow

Ticket: "My monitor is showing a blank screen, but the CPU seems to be running."

Step-by-Step Resolution Flow:

- 1. Check Monitor Power & Status:** Verify that the monitor is actually turned on. Check the LED indicator on the monitor bezel. Is it completely dark (no power), amber/orange (standby/no signal), or white/blue/green (receiving signal but displaying black)?

2. **Restart Graphics Driver:** If the computer is running Windows, instruct the user to press Win + Ctrl + Shift + B. This restarts the graphics driver and can often wake up a blank display.
3. **Verify Physical Connections:** Trace the display cable (HDMI, DisplayPort, VGA) from the back of the monitor to the back of the PC. Unplug and reseat both ends to ensure a secure connection.
4. **Check Connection Port (GPU vs Motherboard):** Ensure the display cable is plugged into the dedicated Graphics Card (GPU) horizontally lower down on the case, and NOT the motherboard I/O panel higher up (if the system has a dedicated GPU).
5. **Verify Input Source:** Use the monitor's physical buttons to open the OSD (On-Screen Display) menu. Ensure the selected input source matches the cable being used (e.g., HDMI 1 vs DisplayPort).
6. **Isolate Components:**
 - Try a different display cable to rule out cable failure.
 - Plug the monitor into another device (like a laptop) to verify the monitor functions.
 - If the monitor works elsewhere, the issue likely resides in the PC's graphics card, RAM, or a failed POST sequence despite the fans spinning.

3. Wireless Mouse Isolation Protocol

Scenario: Unresponsive wireless mouse, but keyboard works fine.

To isolate whether the issue is with the mouse hardware, the USB port, or system settings, I would follow this sequence:

1. **Isolate the Mouse (Power & Hardware):** First, check the mouse itself. Ensure the power switch on the bottom is turned "On". Replace the batteries with fresh ones, or ensure the mouse is fully charged. If the mouse has an optical light on the bottom, verify if it turns on. If it remains completely dead with fresh batteries, the mouse hardware has failed.
2. **Isolate the Connection (Syncing):** For Bluetooth mice, check the OS settings to see if it is still paired. For RF (USB dongle) mice, press the sync/connect button on both the mouse and the receiver if available.
3. **Isolate the USB Port:** Unplug the USB receiver and move it to a different, known-working USB port (preferably on the back of the computer directly connected to the motherboard). If the mouse starts working, the original USB port is faulty.
4. **Isolate the System (Cross-testing):** Plug the USB receiver and mouse into a completely different computer or laptop. If it works on the second computer, the mouse and receiver are fine, meaning the issue lies within the original computer's OS or drivers.
5. **Isolate System Settings/Drivers:** If it only fails on the original computer, open "Device Manager" (Windows). Look under "Mice and other pointing devices" or "Universal Serial Bus controllers" for any devices with a yellow exclamation mark. Uninstall the problematic driver, unplug the receiver, restart the PC, and plug it back in to force a fresh driver installation.

4. Gaming System Overheating Diagnosis

Scenario: A system at a gaming cafe is overheating and shutting down during use.

Here are 3 possible hardware causes and how to check them:

Cause 1: Dust Accumulation Blocking Airflow

Gaming PCs pull in large volumes of air, which brings in dust. Dust clogs case intake filters, radiator fins, and heat sink fins, acting as an insulator and suffocating the system.

How to check: Power down and unplug the system. Open the side panel and visually inspect the internal components. Look for thick layers of dust on the CPU cooler, GPU fans, and front intake mesh. If heavily coated, use compressed air to clean the system thoroughly.

Cause 2: Failed Cooling Fans (Case, CPU, or GPU)

If the fans meant to exhaust hot air or cool the heatsinks fail or seize due to worn bearings, temperatures will rapidly exceed safe thresholds.

How to check: Turn the computer on with the side panel removed. Visually verify that all case fans, the CPU fan, and the GPU fans are actively spinning. Alternatively, use monitoring software like *HWMonitor* or the BIOS to check fan RPM readouts. If a fan is reporting 0 RPM or visibly not spinning, it needs replacement.

Cause 3: Degraded Thermal Paste / Improper Cooler Mounting

Thermal paste sits between the CPU/GPU die and the heatsink to transfer heat. Over time, it can dry out and crack, losing its thermal conductivity. Alternatively, the cooler mounting pressure might have loosened.

How to check: Install diagnostic software (e.g., MSI Afterburner, Core Temp). Run a stress test or launch a demanding game. If the CPU/GPU temperature instantly spikes to 90°C+ the second a load is applied—despite the heatsink being completely clean and the fans spinning at maximum speed—the thermal transfer has failed. The cooler must be unmounted, the old paste cleaned off with isopropyl alcohol, and fresh thermal paste applied.